# v44 model — experimental conditions: input parameters & where they sit in the proliferation parameter space

## A Input parameters per condition

| | SHH | Ptch1 f | MYCN | p16 | p18 | p21 | HHi | EZH2i | HU | CDK4/6i | cyc? |
|---|---|---|---|---|---|---|---|---|---|---|---|
| **GNP +SHH** | 0.5 | 1 | 1 | 0 | **0.4** | 1× | · | · | · | · | ● |
| GNP +HHi | 0.5 | 1 | 1 | 0 | **0.4** | 1× | ✓ | · | · | · | ○ |
| GNP +EZH2i | 0.5 | 1 | 1 | 0 | **0.4** | 1× | · | ✓ | · | · | ● |
| GNP serum-starve | 0.5 | 1 | 1 | 0 | **0.4** | 1× | · | · | · | · | ○ |
| Ptch+/- (P7) | 0.5 | **0.5** | **1.38** | **0.06** | **0.84** | **1.4×** | · | · | · | · | ● |
| **MB** | 0.5 | **0.1** | **2.86** | **0.15** | **1.5** | **2×** | · | · | · | · | ● |
| MB +HHi | 0.5 | **0.1** | **2.86** | **0.15** | **1.5** | **2×** | ✓ | · | · | · | ○ |
| MB +HHi +EZH2i | 0.5 | **0.1** | **2.86** | **0.15** | **1.5** | **2×** | ✓ | ✓ | · | · | ○ |
| MB +CDK4/6i | 0.5 | **0.1** | **2.86** | **0.15** | **1.5** | **2×** | · | · | · | ✓ | ○ |
| MB +HU | 0.5 | **0.1** | **2.86** | **0.15** | **1.5** | **2×** | · | · | ✓ | · | ● |

*context inputs (p16, p18, p21 = CDK-inhibitor tone)* *drug switches*

p16: competitive CDK4/6 brake (raises K_CdRb) · p21: CDK2 inhibitor (kSyP21, ×baseline) · serum-starve k_Cd_trans→0 · CDK4/6i kPhRbCd→0

## B Parameter space: CyclinD1 vs the CDK-inhibitor brake